Option-Only Portfolio Optimization with Greek-Induced Covariance and Conditional Risk Premia

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Abstract

We develop a mean–variance framework for portfolios composed entirely of listed options. Option positions enter as premium-funded, expiring cashflows: premium weights replace the unit budget constraint, covariance is induced by a Greek/state factor map with regularized residual risk, expected returns load on structural risk-premium channels rather than historical contract means, and the allocation solves a conic program whose feasible set carries transaction costs, Greek and stress budgets, margin, and pre-trade liquidity caps. We estimate the model on four option universes—eight equity-option names, the same names plus VIX options, 56 names, and 56 names plus VIX—at a one-million-dollar account size with contract-level costs and whole-contract positions. The two VIX-enabled portfolios earn full-cost net Sharpe ratios of 1.628 and 1.383 (gross 2.010 and 1.675) and beat capped naive option diversification by 1.3 to 2.3 Sharpe points ($p < 0.001$); the no-VIX universes do not clear the same tests. Channel ablations attribute the premium to option carry disciplined by stress and vega budgets. Breadth pays only when contract-level historical means are replaced by shrunken structural forecasts, and displayed option depth confines the strategy to small accounts.

Key results

- `larger+VIX` (56 names + VIX): net Sharpe 1.628, net Sortino 4.377; beats both baselines; rolling out-of-sample Sharpe 1.273.
- `orig+VIX` (8 names + VIX): net Sharpe 1.383, net Sortino 3.551; beats both baselines.
- `larger` (56 names): net Sharpe 0.587, indistinguishable from capped naive diversification (0.596) and below stock Markowitz (0.909).
- `orig` (8 names): positive point estimate (0.778), but displayed option depth cannot support a full book at $1 million NAV.

1. Research question and scope

The paper asks a narrow portfolio question. Can an option-only allocator use the structure of listed options—premium funding, Greeks, volatility exposure, skew, tail insurance, expiry, margin, and displayed liquidity—to build a small-account strategy that survives realistic frictions? In the data studied here the answer is yes for the VIX-enabled variants; Section 6 details the capacity and execution-data limits of that answer.

The negative parts of the result are as important as the positive ones. First, broad option universes do not automatically help. Breadth helps when the mean estimator is made structural and the residual covariance is regularized; otherwise the optimizer uses the extra contracts to overfit historical option means. Second, gross Sharpe is not a trading result. It becomes economically relevant only after spreads, fees, slippage, margin, assignment, and capacity are charged. Third, capacity is a binding market constraint. The tested month-end contracts support a small account at roughly \$1 million NAV, but the same design does not scale mechanically to \$5 million or more.

The contribution is therefore twofold. The theoretical contribution is a compact option-only Markowitz framework: option contracts become premium-normalized cashflows, their covariance is induced by a Greek/state map plus residual risk, and the allocation is a robust conic problem with costs and operational budgets. The empirical contribution is a fully reproducible application to four universes and two baselines, with a specification fixed in advance rather than re-tuned inside validation folds.

The empirical design favors transparency over promotion. Exact panel CBBO is separated from imputed spread rows, capacity-infeasible books are reported as diagnostics rather than headline results, and the stress tests that fail are reported alongside those that pass. The result is stronger because it is narrower: within its stated scope, the locked VIX-enabled specification survives every test we impose. The evidence is a historical backtest rather than a live track record. The paper treats that as a reason to test the specification harder—fixing it in advance and subjecting it to cross-validation, Monte Carlo refitting, and multiple-testing controls—not as a reason to state the supported conclusions more weakly than the data warrant.

1.1. Relation to the literature

The paper sits at the intersection of three literatures. The allocation frame is mean-variance portfolio choice (Markowitz, 1952; Sharpe, 1966), but applied to premium-funded option cashflows rather than budget-constrained asset weights; the breadth and estimation-error mechanics that drive the locked E1 design echo the classical tension between the size of the choice set and the quality of the inputs (Grinold and Kahn, 2000; DeMiguel et al., 2009). The expected-return channels come from the empirical option-return literature: expected option returns and their signs (Coval and Shumway, 2001), delta-hedged gains and the volatility risk premium (Bakshi and Kapadia, 2003), variance risk premia (Carr and Wu, 2009; Bollerslev et al., 2009), index option portfolio evidence (Driessen and Maenhout, 2007; Broadie et al., 2009), direct option-portfolio optimization (Faias and Santa-Clara, 2017), and the embedded-leverage demand that makes cheap convexity systematically expensive (Frazzini and Pedersen, 2022). The validation protocol draws on the backtest-overfitting and multiple-testing literature (White, 2000; Hansen, 2005; Bailey et al., 2017; Bailey and López de Prado, 2014; López de Prado, 2018).

Relative to option-portfolio papers that optimize a small set of index option strategies, the marginal contribution here is operational: contract-level premium accounting, a Greek-induced covariance that prices cross-contract dependence through a state map, pre-trade net liquidity caps inside the feasible set, exact-versus-proxied spread accounting, and VIX futures-option settlement handling, all wrapped in a locked specification that the robustness layer cannot re-tune. None of these pieces is exotic in isolation; the contribution is that the combination is written down, independently verified, and stress-tested as one object.

1.2. Outline

Section 2 develops the theory: premium accounting, the Greek-induced covariance decomposition, the structural expected-return channels with the breadth results that motivate them, and the constrained conic allocation. Section 3 describes the four implementation universes, the locked E1 specification, the transaction-cost stack, and the validation protocol. Section 4 reports gross and net results against both baselines and the mechanism evidence. Section 5 presents the distributional and multiple-testing evidence, including the cross-validation stress analysis. Section 6 states the capacity and execution limits of the claim, and Appendix A collects proofs and solver details. Every figure and table is generated from machine-readable artifacts, and an independent verification harness re-audits the reported numbers, settlement handling, and spread-source policy on each build.

2. Theory: option allocation as funded cashflows

The object of the paper is an option book, not a transformed equity book. An option contract i at decision date t has a mark $C_{i,t}$, multiplier m_i, payoff rule, settlement convention, Greeks, displayed liquidity, and a forecast of conditional option compensation. A portfolio weight w_i is the premium notional committed to that contract divided by NAV. These are *premium weights*. A long option with $w_i = 0.03$ can lose at most three percent of NAV before transaction costs and financing; a short option with $w_i = -0.03$ receives three percent of NAV in premium but carries a future mark-to-market and settlement liability. Cash is the collateral numeraire.

This distinction is the main mathematical reason the option-only problem differs from ordinary stock Markowitz. Stock weights usually satisfy a unit-budget equation. Option premium weights do not. Residual cash, margin collateral, and unpaid short-option losses are separate accounting objects. The correct bridge from theory to implementation is therefore to write option cashflows explicitly, build a Greek-induced covariance for those cashflows, and then constrain the resulting allocation by premium, stress, margin, and liquidity budgets.

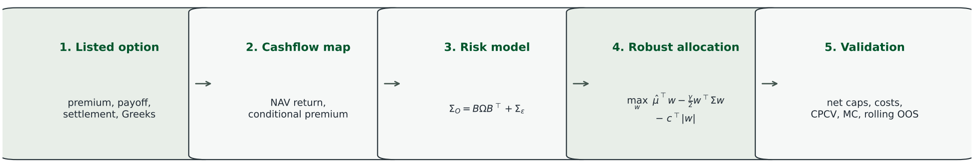

Figure 1: Research pipeline. The theory starts from listed option contracts, turns them into NAV-normalized cashflows, builds Greek-induced covariance and structural expected returns, solves a robust constrained allocation, and then passes the result through execution-cost accounting and validation tests.

2.1. Why ordinary stock Markowitz is insufficient

Ordinary Markowitz (Markowitz, 1952) starts from returns on assets that can be held with weights summing to one. The budget equation does much of the economic work: it fixes scale, makes positive and negative weights interpretable as capital allocation and borrowing, and leaves the covariance matrix as the main risk object. Listed options break that simplification. Premium paid is not the same as capital at risk for a short option; margin required is not the same as premium received; and a low-premium option can carry large delta, gamma, vega, and gap exposure. A variance penalty alone cannot prevent the optimizer from selecting positions that are cheap by premium but expensive by stress loss or market depth.

The right mathematical analogue is therefore not a simple replacement of stock returns by option returns. It is a funded cashflow problem. Premium weights determine the mark-to- market contribution. Collateral and margin constraints determine whether the position can be carried. Greek and stress constraints determine whether local variance misses a bad state. Liquidity caps determine whether the target can be traded at the account size. The portfolio problem becomes a constrained allocation over option cashflow claims, with cash and collateral as the reference asset.

This also explains why the paper keeps a stock Markowitz baseline. If the option-only framework cannot beat ordinary underlying Markowitz after costs and caps, the option structure has not earned its complexity. The stock baseline is not expected to match the option payoff distribution; it is a useful economic hurdle because it asks whether the same underlyings could have been traded more simply. The capped naive option baseline plays the role that $1/N$ plays in the stock literature (DeMiguel et al., 2009): if simple diversification matches the optimizer after the same costs and caps, the estimation machinery has not paid for itself.

2.2. Premium accounting

Let W_t be portfolio NAV at the rebalance date and let $N_{i,t}$ be the signed number of contracts held in option i. With multiplier m_i, define

$$w_{i,t} = \frac{N_{i,t} m_i C_{i,t}}{W_t}.$$

For equity options $m_i = 100$. For VIX options the multiplier convention is handled in the same way through m_i. The one-period option mark return is

$$r_{i,t+1} = \frac{C_{i,t+1} - C_{i,t}}{C_{i,t}}, \qquad \frac{\Delta W_{i,t+1}}{W_t} = w_{i,t} r_{i,t+1}.$$

The multiplier cancels because the weight is already premium-normalized. This is why a cheap out-of-the-money option can have enormous return exposure but still receive a small premium weight. It is also why gross premium, contract concentration, Greek exposure, and stress loss cannot be replaced by a single variance constraint.

Proposition 2.1 (NAV accounting identity)**.** *Let $w_t \in \mathbb{R}^n$ be the option premium-weight vector, r_{t+1} the option mark-return vector, and $r_{f,t}$ the one-period return on residual cash*

$W_t(1 - \mathbf{1}^\top w_t)$. *Ignoring non-option collateral adjustments for notation,*

$$\frac{W_{t+1}}{W_t} - 1 = w_t^\top r_{t+1} + \left(1 - \mathbf{1}^\top w_t\right) r_{f,t}, \qquad \frac{W_{t+1}}{W_t} - 1 - r_{f,t} = w_t^\top (r_{t+1} - r_{f,t}\mathbf{1}).$$

Proof. The option book is worth $\sum_i N_{i,t} m_i C_{i,t+1} = W_t \sum_i w_{i,t}(1 + r_{i,t+1})$. The residual cash account is worth $W_t(1 - \mathbf{1}^\top w_t)(1 + r_{f,t})$. Adding both accounts, dividing by W_t, and subtracting one gives the identity. □

The empirical tables in this paper use a zero-cash benchmark for comparability across strategy variants. Proposition 2.1 is the conversion rule if a reader wants excess-of-cash returns instead. Positive collateral yield would add income to cash-rich long-option books and raise the hurdle for excess Sharpe ratios. The comparison between the four option variants and the two baselines remains internally consistent because the same convention is applied throughout.

Two details matter for implementation. First, a long option cannot lose more than its premium, but a short option can lose far more than the premium it collected. The premium weight w_i is therefore an accounting weight, not a complete risk measure. Second, every result in this paper is reported for a whole-contract book. The maximum-Sharpe solver returns continuous premium weights, but each weight is converted to a signed integer number of contracts before the book is scored, so the Sharpe ratios, Sortino ratios, cost figures, and ledgers below all describe integer positions at the standard m_i multiplier. Remark 2.2 states the conversion rule.

Remark 2.2 (Whole-contract rounding). A premium weight w_i implies a continuous contract count $N_i = w_i W_t/(m_i C_{i,t})$. The traded book rounds N_i to the nearest integer and, wherever a per-contract liquidity cap binds, steps the magnitude down by one contract so that the realized weight $\widehat{w}_i = N_i m_i C_{i,t}/W_t$ never exceeds the cap. All four option books and both capped naive baselines are scored on these realized integer weights. At \$1 million NAV the adjustment is immaterial for liquid large-cap legs but is not negligible for high-priced or thin contracts, where the continuous solution sits exactly at its cap. Because a cap-binding leg can only round down, whole-contract execution slightly lowers deployed gross premium and net cost drag relative to the continuous book rather than adding rounding noise; a floor-toward-zero rule leaves the ranking unchanged with somewhat larger de-leveraging.

2.3. Greek-induced covariance

The option return map comes from a local state expansion. For an equity option on underlying u, write $\Delta S_u = S_{u,t+1} - S_{u,t}$ and $R_u = \Delta S_u / S_{u,t}$. The mark change is decomposed as

$$\Delta C_i = \Delta_{i,t}^{BS} \Delta S_u + \frac{1}{2}\Gamma_{i,t}(\Delta S_u)^2 + \nu_{i,t}\Delta\sigma_{i,t} + \Theta_{i,t}\Delta t + \eta_{i,t+1}.$$

The remainder $\eta_{i,t+1}$ is defined to make this an identity at the rebalance horizon. It absorbs vanna, volga, smile movement, early-exercise premium changes, stale marks, and all higher-order terms not separately modeled. The paper does not require Black–Scholes–Merton to be the true global pricing model for American single-name options. It uses the weaker local fact that option marks can be projected onto spot, curvature, volatility, and time channels (Black and Scholes, 1973; Merton, 1973).

Dividing by $C_{i,t}$ gives premium-normalized state exposure,

$$r_{i,t+1} = b^{\Delta}_{i,t} R_{u,t+1} + b^{\Gamma}_{i,t} R^2_{u,t+1} + b^{\nu}_{i,t} \Delta\sigma_{i,t} + \theta^*_{i,t} + \epsilon_{i,t+1},$$

where

$$b^{\Delta}_{i,t} = \frac{\Delta^{BS}_{i,t} S_{u,t}}{C_{i,t}}, \quad b^{\Gamma}_{i,t} = \frac{\Gamma_{i,t} S^2_{u,t}}{2C_{i,t}}, \quad b^{\nu}_{i,t} = \frac{\nu_{i,t}}{C_{i,t}}, \quad \theta^*_{i,t} = \frac{\Theta_{i,t}\Delta t}{C_{i,t}}.$$

These coefficients are option-return exposures, not dollar Greeks. Their magnitudes can be large because the option premium can be small. That feature is economically correct: cheap convexity is precisely where leverage, spread costs, and liquidity scarcity enter.

For VIX options the traded anchor is not equity spot but the matched VX futures curve. Using Black's futures-option logic (Black, 1976), a VIX option j with matched forward $F^{VX}_{T_j,t}$ enters through

$$r_{j,t+1} = b^{\Delta,VX}_{j,t} R^{VX}_{T_j,t+1} + b^{\Gamma,VX}_{j,t} (R^{VX}_{T_j,t+1})^2 + b^{\nu,VIX}_{j,t} \Delta\sigma^{VIX}_{j,t} + \theta^*_{j,t} + \epsilon_{j,t+1}.$$

VIX, VVIX, and term-structure indices are useful state variables, but VX futures and VIX options are the traded volatility instruments. Exact VRO/SOQ settlement is required before VIX option expiry P&L can be promoted from settlement proxy to listed-settlement evidence.

Stacking all contracts gives the conditional factor model

$$r_{O,t+1} = \mu_{O,t} + B_t f_{t+1} + \varepsilon_{t+1}, \qquad \Sigma_{O,t} = B_t \Omega_t B_t^\top + \Sigma_{\varepsilon,t}.$$

The factor vector f_{t+1} collects underlying returns, centered squared returns, at-the-money implied-volatility shocks, skew and tail-slope shocks, and VX-forward return and curvature shocks. The matrix B_t maps contract-level Greeks into those factors. The residual covariance is not optional. It carries basis risk, smile risk, marking error, and all option return variation not spanned by the Greek map.

Assumption 1 (Projection residual)**.** Let $\Omega_t = \mathrm{Var}_t(f_{t+1}) \succ 0$, and define

$$B_t = \mathrm{Cov}_t(r_{O,t+1}, f_{t+1})\Omega_t^{-1}, \qquad \varepsilon_{t+1} = r_{O,t+1} - \mu_{O,t} - B_t f_{t+1}.$$

Then $\mathrm{Cov}_t(f_{t+1}, \varepsilon_{t+1}) = 0$ by construction.

Proposition 2.3 (Covariance identity)**.** *Under Assumption 1,*

$$\mathrm{Var}_t(r_{O,t+1}) = B_t \Omega_t B_t^\top + \Sigma_{\varepsilon,t}, \qquad \Sigma_{\varepsilon,t} := \mathrm{Var}_t(\varepsilon_{t+1}),$$

and both terms are positive semidefinite.

Proof. Substitute the projection decomposition into the variance and use $\mathrm{Cov}_t(f_{t+1}, \varepsilon_{t+1}) = 0$. Positive semidefiniteness follows because each term is a covariance matrix or a linear image of one. □

The empirical estimator follows this identity but keeps the implementation conservative. The factor covariance is estimated from training-window state variables and shrunk. The residual covariance is estimated from option residuals on an unbalanced panel, projected onto the PSD cone when pairwise moments create small negative eigenvalues, and then regularized. The locked E1 model uses diagonal residual covariance because the dense residual matrix was an overfitting channel in the broad universe.

2.4. State factors and residual risk

The factor map has an economic interpretation. Delta exposure routes option returns into underlying equity returns. Gamma exposure routes them into realized quadratic movement. Vega exposure routes them into implied-volatility shocks. Skew and tail variables proxy where crash insurance and wing convexity are being repriced. VX-forward exposure lets VIX options behave as options on the traded volatility futures curve rather than as options on an untraded spot index.

The residual term is where the model acknowledges its limits. If option marks were perfectly spanned by spot, variance, volatility, skew, and VIX-forward factors, $\Sigma_{\varepsilon,t}$ would be zero. It is not. Residual risk captures early exercise, discrete dividends, surface deformations, quote noise, strike/maturity mismatch, and the fact that a monthly rebalance is not an infinitesimal Ito step. The model can use Greeks without pretending that the Greek map is complete.

The diagonal residual choice in E1 should be read in this spirit. It is not a claim that residual risks are truly independent. It is a regularization decision. In a broad option universe, a dense residual covariance matrix estimated from a short and unbalanced panel can create false diversification and unstable inverse-covariance directions. Diagonal residual covariance keeps contract-specific unexplained risk while refusing to let noisy off-diagonal residual estimates dominate the optimizer.

2.5. Expected option compensation

Greek covariance alone cannot create a Sharpe ratio. The optimizer needs a conditional mean vector. The economic decomposition is

$$\widehat{\mu}_{i,t} = b^{\Delta}_{i,t}\widehat{\lambda}^{S}_{t} + b^{\Gamma}_{i,t}\widehat{\lambda}^{var}_{t} + b^{\nu}_{i,t}\widehat{\lambda}^{vol}_{t} + b^{skew}_{i,t}\widehat{\lambda}^{skew}_{t} + b^{tail}_{i,t}\widehat{\lambda}^{tail}_{t} + \theta^{*}_{i,t}.$$

Delta loads on equity compensation. Gamma and theta jointly encode the realized-versus-implied variance wedge. Vega loads on volatility-risk compensation. Skew and tail terms represent compensation for crash insurance and convex volatility exposure, which the option-return literature documents in multiple forms (Coval and Shumway, 2001; Bakshi and Kapadia, 2003; Carr and Wu, 2009; Bollerslev et al., 2009).

The implemented signal is simpler than the general display. It combines carry, a damped delta channel, a vega-routed implied-minus-realized volatility wedge, and skew/tail characteristics. Cheap convexity is treated as systematically expensive rather than anomalous, consistent with the embedded-leverage evidence of Frazzini and Pedersen (2022). The locked E1 version then removes the raw historical option mean entirely and shrinks the structural forecast toward zero by 0.75. This is not an aesthetic choice. It is the central empirical lesson of the breadth experiments, and it is the option-panel version of the classical breadth-versus-input-quality tension (Grinold and Kahn, 2000): the historical mean is useful when the option universe is small and the training ratio T/n is healthy, but it becomes a ranking-noise generator when the optimizer can choose among many correlated contracts.

Proposition 2.4 (Historical mean error and breadth)**.** *Let the training window contain T monthly observations of n contracts, and let the sample-mean errors $e_i = \widehat{\mu}_i^{\text{hist}} - \mu_i$ be mean zero and sub-Gaussian with variance proxies σ_i^2/T, $\bar{\sigma} = \max_i \sigma_i$. Then*

$$\mathbb{E}\Big[\max_i |e_i|\Big] \;\leq\; \bar{\sigma}\sqrt{\frac{2\log(2n)}{T}},$$

and for any feasible set $\mathcal{C} \subseteq \{w : \|w\|_1 \leq g\}$ the plug-in optimizer's expected objective inflation obeys $\mathbb{E}\big[\sup_{w\in\mathcal{C}} e^\top w\big] \leq g\,\bar{\sigma}\sqrt{2\log(2n)/T}$. The allocation chosen under the noisy mean therefore sacrifices at most $2g\,\bar{\sigma}\sqrt{2\log(2n)/T}$ of true objective value in expectation, and this worst-case loss grows with n at fixed T.

Corollary 2.5 (Bias–variance crossover)**.** *Suppose the structural forecast has a deterministic specification bias bounded by $\beta = \|\mu^{\text{struct}} - \mu\|_\infty$ that does not grow with n. Comparing worst-case objective losses, the historical mean is preferred only when $\bar{\sigma}\sqrt{2\log(2n)/T} < \beta$, that is, when $n < \frac{1}{2}\exp\big(T\beta^2/2\bar{\sigma}^2\big)$. With T fixed, a historical option mean can therefore help at small n and hurt at broad n.*

Proofs are in Appendix A.2. The sub-Gaussian model is a stylization: option panels add correlation, missingness, wide tails, and changing contract composition, all of which raise the effective error at a given (n, T) rather than lower it. The bound therefore understates the practical penalty of ranking many nearly substitutable contracts by noisy estimated means.

The crossover is visible in the artifacts. At eight names, dropping the historical mean hurts the no-VIX gross Sharpe, moving it from 0.842 to 0.725. In the broad universe, keeping the historical mean is the dominant source of optimizer failure: structural-only forecasts and stronger shrinkage are what allow the 56-name and 57-name books to become viable portfolios. The theory does not say the historical mean is always bad. It says the mean estimator must respect the ratio between sample length and choice set.

Remark 2.6 (Shrinkage to zero as a posterior mean). The 0.75 shrink-to-zero factor has a standard Bayesian reading. If the true premium satisfies $\mu_i \sim N(0, \tau^2)$ and the structural forecast is unbiased with noise $\widehat{\mu}_i^{\text{struct}} \mid \mu_i \sim N(\mu_i, s^2)$, the posterior mean is $(1-\kappa)\,\widehat{\mu}_i^{\text{struct}}$ with $\kappa = s^2/(s^2+\tau^2)$. E1's $\kappa = 0.75$ corresponds to a prior signal-to-noise ratio $\tau^2/s^2 = 1/3$: the forecast is treated as three parts noise to one part signal. Because the structural blend itself carries a weight of 0.75, the effective multiplier on the raw structural signal is $(1-0.75)\times 0.75 = 0.1875$. The same regularize-toward-a-simple-target logic is applied to the covariance (Ledoit and Wolf, 2004). Both parameters were fixed in the earlier diagnostic phase and locked before the validation runs; neither is re-tuned inside folds.

2.6. From economic premia to the E1 signal

The structural mean used by E1 can be summarized as four channels. The carry channel is $\theta^*_{i,t}$, the premium-normalized theta contribution under unchanged state variables. The delta channel applies a damped forecast of the underlying's training-window return to $b^{\Delta}_{i,t}$. The volatility channel routes the implied-minus-realized gap through vega rather than double-counting it through both gamma and theta. The skew/tail channel gives insurance-like wings a negative expected return for long positions and a positive expected return for short positions, subject to stress and liquidity caps.

This construction favors stable economic signs over flexible statistical fit. It does not estimate a separate mean for every contract from that contract's past returns. It asks whether a contract is expensive or cheap relative to its structural exposures, then shrinks the result heavily toward zero. That is why the locked E1 result is interpretable: the broad-universe improvement comes from removing an identified high-variance estimator, not from adding an opaque machine-learning layer.

The no-free-lunch constraint is that structural premia must be paid for in risk budgets. Shorting skew or volatility can earn compensation precisely because those positions lose in crash and volatility-spike states. The optimizer is therefore never allowed to view the mean in isolation. A contract with attractive expected carry can still be excluded by stress loss, vega exposure, beta exposure, margin, concentration, or liquidity.

2.7. Robust conic allocation

Given $\widehat{\mu}_t$ and $\widehat{\Sigma}_{O,t}$, the headline allocation is

$$\max_{w \in \mathcal{C}_t} \widehat{\mu}_t^\top w - \frac{\gamma}{2} w^\top \widehat{\Sigma}_{O,t} w - c_t^\top |w|. \tag{P}$$

The cost vector c_t includes bid-ask spread, explicit fees, slippage, borrow proxy, margin drag, assignment penalty, and capacity/impact costs; charging linear costs inside the objective rather than after it follows the cost-aware trading literature (Gârleanu and Pedersen, 2013). The feasible set $\mathcal{C}_t$ imposes premium, Greek, beta/VIX, stress-loss, margin/collateral, concentration, and liquidity constraints. The formulation is robust in the conservative-inputs sense of conic robust portfolio selection (Ben-Tal and Nemirovski, 1998; Goldfarb and Iyengar, 2003): shrunk means, regularized covariance, and hard budgets replace point estimates rather than an explicit uncertainty ellipsoid. In implementation the maximum-Sharpe solver is the ratio-form proxy for this robust conic allocation, with regularized inputs and hard pre-trade caps.

The unconstrained option-only geometry is a useful warning. If $\widehat{\Sigma}_{O,t} \succ 0$ and $\widehat{\mu}_t \neq 0$, the maximum-Sharpe direction is proportional to $\widehat{\Sigma}_{O,t}^{-1}\widehat{\mu}_t$. Unlike a stock-only problem, there is no required $\mathbf{1}^\top w = 1$ budget that bends the efficient frontier into the usual hyperbola.

Proposition 2.7 (Option-only frontier is a ray)**.** *For a target option mean $m > 0$, the minimum-variance premium-weight vector in the unconstrained problem is*

$$w^*(m) = \frac{m\,\Sigma^{-1}\mu}{\mu^\top \Sigma^{-1}\mu}, \qquad \sigma(m) = \frac{m}{\sqrt{\mu^\top \Sigma^{-1}\mu}}.$$

Thus the unconstrained efficient frontier in (σ, m) space is a ray through the origin, and the operational constraints choose the implementable segment.

Proof. The Lagrangian for $\min_w w^\top \Sigma w$ subject to $\mu^\top w = m$ yields $2\Sigma w = \lambda\mu$. The constraint pins $\lambda/2 = m/(\mu^\top \Sigma^{-1}\mu)$. Substitution gives the displayed variance. □

The most important operational constraint in this paper is the net liquidity cap

$$|w_i| \le \bar{w}_{i,t}, \qquad \bar{w}_{i,t} = \min\left\{ 0.18,\ \frac{X\,\widehat{V}_{i,t}^{train}\,M_{i,t}\,100}{\mathrm{NAV}} \right\}.$$

Here $X = 0.05$, $M_{i,t}$ is the option mark, and $\widehat{V}_{i,t}^{train}$ is a training-window volume statistic. The cap is point-in-time because it uses only train-window volume. It is net because it binds the economic position w_i, not the positive and negative solver legs separately.

Proposition 2.8 (Net caps and representation invariance)**.** *A cap on $|w_i|$ is invariant to algebraic representation of a contract position. A cap on separate long and short legs is not representation-invariant and can create pseudo-cash artifacts.*

Proof. The economic position is $w_i = w_i^+ - w_i^-$. The bound $|w_i| \le \bar{w}_i$ depends only on that net exposure. Separate bounds on w_i^+ and w_i^- allow both legs to be positive while the net position is small, so a solver can consume displayed capacity in offsetting long/short pairs without taking the intended exposure. □

Proposition 2.9 (Costs and tighter feasible sets)**.** *If costs rise componentwise or $\mathcal{C}_t$ is tightened, the optimal value of (P) weakly decreases. Without premium, collateral, liquidity, and stress constraints, option exposure can be mechanically scaled and the problem is not an implementable allocation rule.*

Proof. For any fixed feasible w, increasing c_t weakly lowers the objective. Maximizing over a subset cannot improve the supremum. Without scale constraints, a positive objective direction can be levered until an omitted real-world budget binds. □

These propositions delimit what the empirical sections can and cannot show. The paper is not trying to show that any option signal can be made to work with enough leverage. It asks whether a locked, regularized, cap-constrained option allocation survives when the market frictions that matter for small-account trading are imposed before performance is reported.

2.8. Explicit constraint set

A useful way to read the feasible set is

$$\begin{aligned}\mathcal{C}_t = \{w \in \mathbb{R}^n :\ & \|w\|_1 \le g, \qquad \sum_{i:w_i<0} |w_i| \le s, \qquad |w_i| \le \bar{w}_{i,t}\ \forall i,\\ & \sum_{i:u(i)=u} |w_i| \le g_u\ \forall u, \qquad |a_{k,t}^\top w| \le b_k\ \forall k,\\ & \ell_{j,t}^\top w \ge -m_j\ \forall j, \qquad \mathrm{Margin}_t(w) \le M_t\}.\end{aligned}$$

The first line controls gross premium, short premium, and contract-level liquidity. The second controls underlying concentration and Greek/beta/VIX budgets. The third controls scenario stress loss and collateral/margin. The vectors $a_{k,t}$ include NAV-normalized delta, beta, gamma, equity vega, VIX vega, and other exposure rows. The stress rows $\ell_{j,t}$ reprice the book under adverse spot, volatility, and VIX shocks.

The quadratic risk term can be handled as a second-order cone when the problem is written in risk-budget or utility form. The absolute-cost term $c_t^\top |w|$ is linearized by standard positive and negative parts. The liquidity cap must bind the net variable, as Proposition 2.8 states. This is why the implementation detail that fixed the split-leg pseudo-cash artifact belongs in the paper's theory section rather than only in a code note: the algebraic representation of a conic solver must preserve the economic position. Appendix A records the corresponding maximum-Sharpe, second-order-cone, and estimator-repair details without interrupting the empirical narrative.

2.9. Algorithmic summary

The locked E1 procedure can be written as a short algorithm.

1. Build the eligible option universe at the decision date and compute premium- normalized Greeks, marks, volumes, and settlement labels.
2. Estimate state-factor covariance from the training window; estimate residual risk from option residuals; project to PSD where needed and use diagonal residual covariance.
3. Form the structural expected-return vector from carry, delta, volatility, skew/tail, and VIX channels; set historical option mean weight to zero; shrink the mean 75% toward zero.
4. Apply N-scaled covariance shrinkage and train-window net liquidity caps with $X = 0.05$.
5. Solve the constrained option allocation and evaluate gross, full-cost net, capacity, and deployability metrics.

The important methodological point is that all fold-specific and rolling refits rebuild the representative contracts, mean/covariance estimates, and liquidity caps from the fold's training window. Liquidity caps use training-window volume only. The strategy is locked before validation, so robustness checks do not get to choose a different estimator after seeing test performance.

3. Empirical application

The empirical application turns the clean conic allocation of Section 2 into an exchange-realistic book by layering three data-driven frictions onto the idealized problem. First, every position is priced through a full transaction-cost stack—spread crossing, fees, slippage, borrow, margin drag, and assignment penalty—built from historical market-hours CBBO quotes rather than a flat default spread. Second, each contract carries a volume-aware pre-trade liquidity cap that sizes it to a fixed fraction $X = 0.05$ of its displayed training-window volume, so the optimizer can only take exposure the market actually showed (the cap $\bar{w}_{i,t}$ of Section 2). Third, the continuous premium weights are rounded to whole option contracts at the standard m_i multiplier before any performance number is computed (Remark 2.2). The theory says which exposures to hold; these three layers constrain the book to the spreads, depth, and contract granularity that the historical data actually recorded, and every headline result is reported only after all three are imposed. The subsections below give the four universes, the cost stack and its spread sourcing, and the validation protocol in turn.

3.1. Four universes and locked model

The empirical design uses four implementation universes. `orig` is the original eight equity-option names: AAPL, AMZN, GOOGL, JPM, META, MSFT, NVDA, and TSLA. `orig+VIX` adds VIX options. `larger` expands the equity-option universe to 56 names. `larger+VIX` adds VIX to that broader set. The four variants are not four different optimizers. They are four applications of the same locked E1 model.

Config	Universe	Spread input	Status	E1 net	Naive net	Stock Markowitz
orig	8 equities	exact panel CBBO	capacity diagnostic	0.778	0.365	0.909
orig+VIX	8 equities + VIX	exact panel CBBO + inferred CBBO proxy	research candidate	1.383	-0.890	0.909
larger	56 equities	exact panel CBBO + inferred CBBO proxy	mixed	0.587	0.596	0.909
larger+VIX	56 equities + VIX	exact panel CBBO + inferred CBBO proxy	research candidate	1.628	0.293	0.909

Table 1: Four empirical scenarios. The stock baseline is ordinary underlying Markowitz. The naive option baseline is the better of capped equal-premium and capped equal-risk option books under the same cost and cap stack.

The locked E1 specification uses structural-only means, diagonal residual covariance, N-scaled covariance shrinkage, shrinkage-to-zero 0.75, net liquidity caps with $X = 0.05$, and \$1 million NAV. It was chosen because the earlier breadth diagnostic found two failure modes: the broad-universe historical mean overfit the contract ranking, and the old post-hoc capacity penalty let the optimizer choose positions that could not be pre-trade deployed. Neither fix alone was sufficient. The combined specification is therefore the locked research design, not an in-test knob search.

The contract construction is representative rather than high-frequency. Each rebalance date maps the eligible option panel into month-end representative contracts and then holds the resulting static book over the next period in the headline split. This choice makes the backtest auditable: the research question is whether a monthly, capacity-aware option allocation works after costs, not whether intraday execution timing can rescue a fragile signal. It also makes the capacity result sharper. If month-end representative contracts cannot support the desired NAV under a five-percent participation cap, the strategy should not be promoted by assuming invisible intraday depth.

The two baselines are chosen for interpretability. The first is ordinary Markowitz on the underlying stocks. It asks whether option structure adds anything beyond an equity portfolio on the same economic names. The second is a capped naive option book: equal premium or equal risk after applying the same liquidity caps and costs. It asks whether the Greek/mean/covariance theory beats a mechanically diversified options sleeve. The paper's claim requires beating both baselines while remaining deployable under the cap budget.

The two baselines serve different failure modes. Stock Markowitz is a simplicity benchmark: if the option book cannot beat a portfolio of underlying shares, the option engineering may be unnecessary. The capped naive option book is a structure benchmark: if equal-premium or equal-risk option diversification performs as well as E1, then the Greek covariance and structural mean are not adding much. The broad no-VIX result fails this second test because its E1 Sharpe is essentially tied with capped naive options.

3.2. Cost stack and spread sourcing

Every headline E1 result uses the corrected cost stack: spread crossing, fees, slippage, borrow proxy, margin drag, assignment penalty, and capacity/impact costs. The most consequential design decision in that stack is spread sourcing. A flat 10% default spread is not a realistic representation of liquid large-cap options and materially deflates large parts of the option universe. The corrected stack uses exact historical market-hours panel CBBO where available and an inferred CBBO proxy only where matched broad historical quotes are missing.

The full-cost return for a book is therefore not simply gross return minus one spread number. Costs enter through several channels. Spread and fee costs scale with the target option premium traded. Slippage is a further execution allowance. Borrow and margin drag represent the cost of carrying short or collateral-intensive exposure. Assignment penalty is a fail-closed proxy for short American-option exercise risk. Capacity/impact costs penalize positions that try to consume too much displayed volume. The corrected headline results use these costs together; no row is promoted from a gross-only statistic.

Spread sourcing shapes what the results can support because OPRA is the consolidated U.S. options quote and last-sale system,[1] and Cboe sells historical option NBBO snapshots with size, including 15:45 snapshots intended to be more representative than end-of-day liquidity.[2] Matched broad historical market-hours NBBO/CBBO is generally not free. Therefore this paper separates exact evidence from calibrated execution sensitivity. The baseline eight equity-option rows use exact historical panel CBBO. VIX rows and missing broad-name rows use a point-in-time inferred CBBO proxy calibrated from the liquid CBBO surface. Proxy rows are not final execution truth.

Config	Exact CBBO rows	Proxy rows	Proxy share	Median spread
orig	5777	0	0.000	0.019
orig+VIX	5777	536	0.085	0.022
larger	5777	23740	0.804	0.019
larger+VIX	5777	24276	0.808	0.020

Table 2: Spread-source coverage in the corrected cost stack. The proxy share is row-weighted across cost rows, not an exchange-wide liquidity statistic.

The imputation ladder is intentionally conservative in wording even when the calibrated spreads are economically plausible. For the eight equity names, the cost rows are exact historical market-hours CBBO. For `orig+VIX`, the equity-option rows remain exact and the VIX rows are inferred from the liquid-option surface. For broad rows, added names without matched historical CBBO use the same point-in-time proxy. This is enough to replace the stale 10% default in research results, but it is not enough to certify live tradability. The next data upgrade is paid or broker-provided historical market-hours NBBO/CBBO with displayed size matched to every backtest decision date.

The table also explains why the paper gives stronger evidentiary weight to the baseline eight equity rows than to the broad spread rows. `orig` has zero proxy spread rows. `orig+VIX` has exact equity-option rows and proxied VIX rows. The broad variants use proxy rows for most added-name cost observations. That does not invalidate the broad exercise: the proxy

[1] Options Price Reporting Authority: `https://www.opraplan.com/`.
[2] Cboe DataShop Option EOD Summary: `https://datashop.cboe.com/option-eod-summary`.

is point-in-time and calibrated from observed liquid-option CBBO. It does, however, prevent a final execution claim. The broad result is best read as: under realistic liquid-option spread assumptions, the optimizer survives; under exact historical broad NBBO/CBBO, this still needs to be confirmed.

3.3. Validation protocol

The application is static-weight at the headline split and refit-based in the robustness layer. The four variants are evaluated at $\$1$ million NAV and $X = 0.05$ participation because larger NAVs hit the option market's depth constraint rather than an optimizer constraint. Primary deployable status requires the hard cap budget to support the static book. Capacity-infeasible rows are reported only as diagnostics even when their point estimate is positive.

The robustness layer uses 12 chronological groups, 66 CPCV splits run in both a full-history design and a claim-window design restricted to the post-2020 evaluation months (Section 5), 78 total CV splits in the PBO summary (Bailey et al., 2017; López de Prado, 2018), one-month purge/embargo, Monte Carlo resampled histories built from circular-block bootstrap draws (Politis and Romano, 1994), Monte Carlo refit stability, repriced synthetic option universes with a historical-cost overlay, circular-block bootstrap path simulations, GARCH/EGARCH-style volatility-path diagnostics (Nelson, 1991) with fallback when the EGARCH sample is too short, drawdown breach simulations, multiple-testing/reality-check inference, and rolling 36-month OOS refits. These checks are designed to attack the result from multiple angles rather than to create new degrees of freedom.

The purge and embargo rule is simple: one month is dropped around fold boundaries so option labels whose payoff windows overlap the test block do not leak into training. Lemma A.3 in the appendix records why one month is sufficient: the tested contracts' decision-to-expiry spans are at most about 44 calendar days, while the verifier-measured minimum gap between the last usable training month and the first test month across all folds is 59 days. The CPCV exercise is distributional, not tradable, because some training sets can include future calendar months relative to a tested block. Rolling OOS is closer to a trading protocol because it refits through time. Monte Carlo refit stability asks whether the procedure selects similar risk-adjusted books under alternate training histories. Repriced synthetic paths ask whether the strategy depends too much on one realized option surface path. Each diagnostic attacks a different source of false confidence.

3.4. Performance metrics and deployability

Performance is reported in full-cost net Sharpe and Sortino units unless the text explicitly says gross. With monthly returns $R_1, \ldots, R_T$, the annualized Sharpe ratio uses

$$\widehat{\mathrm{SR}} = \frac{12\,\bar{R}}{\sqrt{12}\,\widehat{\sigma}_R}.$$

The Sortino ratio replaces $\widehat{\sigma}_R$ with downside semideviation around the zero-return threshold (Sortino and Price, 1994); drawdown-based and distribution-based complements (Magdon-Ismail and Atiya, 2004; Keating and Shadwick, 2002) appear in the artifact ledgers. Sharpe point estimates on 60 monthly observations carry wide sampling bands (Lo, 2002), which is why Section 5 reports interval and difference-test evidence rather than point estimates alone.

These ratios are convenient summaries, but they are not sufficient for a positive verdict. A strategy can have a positive Sharpe and still be rejected if the cap budget cannot deploy the book, if the result is tied with a simpler baseline, or if the spread source is too weak for the claim being made.

Deployability is a separate requirement. A row is deployable only when the hard net-cap problem can allocate the intended NAV without relying on relaxed capacity. If the sum of contract-level caps is below one NAV, the optimizer cannot build a full book at the specified account size. Such rows stay in the paper because they diagnose the mathematics, but they do not become headline trading candidates. This is why the `orig` no-VIX row is marked capacity diagnostic even though its full-cost net Sharpe is positive.

The same logic applies to baseline dominance. A row that is positive but tied with capped naive options does not validate the full theory. It may show that option exposure has a premium, but it does not show that Greek-induced covariance, structural means, and conic constraints improve on simple option diversification. The broad no-VIX row falls into this category.

3.5. Artifact discipline

The empirical claim is backed by generated artifacts rather than hand-entered numbers. The headline figures and tables are rebuilt from machine-readable robustness and cost artifacts. The spread-source coverage table is built from the cost ledger. The validation distributions are built from CPCV, Monte Carlo, refit, and rolling OOS outputs. The paper verifier checks page quality, scope statements, artifact schemas, spread-source policy, VIX settlement handling, and cap point-in-time behavior.

This discipline matters because three design errors identified during the research process would otherwise be invisible in a summary table: charging default spreads to names with observable quotes, penalizing capacity after optimization instead of constraining it before, and a split-leg liquidity formulation that wastes displayed capacity as pseudo-cash. The final empirical object corrects all three; it is the locked E1 candidate with corrected spreads, full costs, and net caps.

The artifact ledger also protects against a common failure in option research: inadvertently mixing data vintages. Liquidity caps use training-window volume only. Fold-specific models rebuild the contract set and risk estimates from the fold's training window. Broad spread proxy rows are point-in-time calibrated rather than filled from future quote observations. These controls do not make the backtest live trading evidence, but they keep the simulation internally auditable.

4. Results: four variants and two baselines

4.1. Headline comparison

The results concentrate in the VIX-enabled universes: those books clear every practical screen, while the no-VIX books do not. `larger+VIX` is the strongest candidate, with net Sharpe 1.628 and net Sortino 4.377. `orig+VIX` also beats both baselines, with net Sharpe 1.383 and net Sortino 3.551. `larger` without VIX is positive but essentially tied with the capped naive option book and below stock Markowitz. `orig` without VIX has positive point estimates but fails the deployability test because the hard liquidity-cap budget is below one

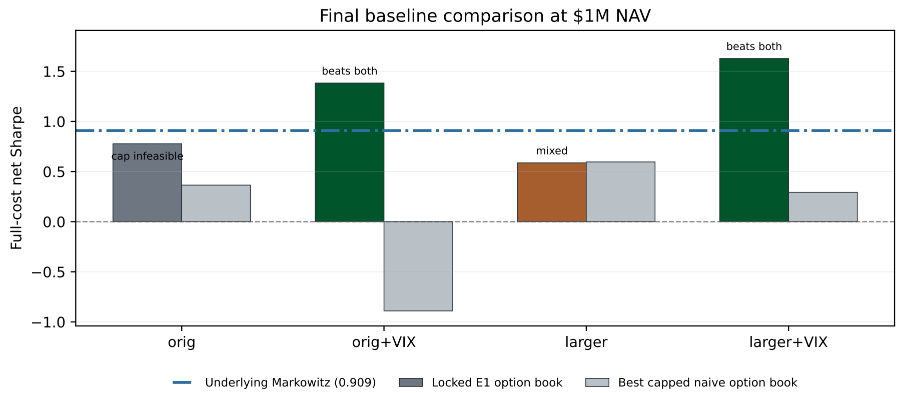

Figure 2: Final four-variant comparison at \$1 million NAV. The VIX-enabled locked E1 books beat both baselines. The no-VIX books are weaker: broad no-VIX is mixed, and original no-VIX is a capacity diagnostic.

Config	Verdict	Gross Sharpe	Net Sharpe	Net Sortino	Naive Sharpe	Stock Sharpe	Rolling Sharpe	MC resample p05	MC refit p05
orig	capacity diagnostic	0.975	0.778	1.783	0.365	0.909	0.585	0.199	0.570
orig+VIX	pass	1.675	1.383	3.551	-0.890	0.909	1.033	0.873	1.178
larger	mixed	0.847	0.587	1.298	0.596	0.909	0.480	-0.001	0.537
larger+VIX	pass	2.010	1.628	4.377	0.293	0.909	1.273	1.096	1.360

Table 3: Compact numeric scoreboard behind Figure 2. Metrics are full-cost net unless a column is explicitly labeled gross.

NAV.

Gross and net results are reported side by side so the reader can see exactly what the cost stack removes. The gross-to-net Sharpe pairs are $0.975 \to 0.778$ for `orig`, $1.675 \to 1.383$ for `orig+VIX`, $0.847 \to 0.587$ for `larger`, and $2.010 \to 1.628$ for `larger+VIX`. The corrected cost stack therefore removes between 0.20 and 0.38 Sharpe points across the four books, and every headline claim in the paper is stated on the net side of that gap.

Two comparisons matter. Against underlying stock Markowitz, the VIX-enabled option books earn a higher net Sharpe in this simulation while still being subject to explicit option costs and liquidity caps. Against capped naive options, the VIX-enabled E1 books earn a large optimizer edge: `orig+VIX` moves from a negative naive option Sharpe to a positive E1 Sharpe, and `larger+VIX` beats the capped naive option benchmark by more than one Sharpe point. The no-VIX broad row does not earn this interpretation because 0.587 for the optimizer versus 0.596 for capped naive is not an economically meaningful edge in either direction.

The Sortino ratios tell the same story from downside variation. `larger+VIX` reaches net Sortino 4.377, and `orig+VIX` reaches 3.551. The no-VIX rows are materially weaker: `orig` has Sortino 1.783 but fails capacity, while `larger` has Sortino 1.298 and does not beat the stock baseline. The downside metrics therefore reinforce the ranking rather than changing it.

It is also important that the result is not driven by treating relaxed or infeasible rows as deployable candidates. A positive point estimate is not sufficient. The candidate must survive the hard net liquidity cap, full-cost accounting, and baseline comparisons. This is why the original no-VIX row is not promoted even though its E1 Sharpe is positive, and why the broad no-VIX row is marked mixed even though it is deployable.

4.2. Mechanism: breadth, mean regularization, and caps

The result is not simply that more names improve the opportunity set. Breadth helps only after the estimator and liquidity formulation are fixed together. The dominant estimator mistake was the historical mean. In the larger+VIX universe, dropping it and using structural-only forecasts recovers much of the gross Sharpe that the broad paper configuration lost. Diagonal residual covariance and N-scaled shrinkage then remove a second overfitting channel. The pre-registered combination reaches a gross Sharpe of 2.010 for the broad VIX-enabled book and produces the net Sharpe 1.628 after the corrected cost stack.

Arm	orig	orig+VIX	larger	larger+VIX
Full E1	0.778	1.383	0.587	1.628
No carry	0.774	0.463	0.574	0.483
No delta	0.370	1.479	0.672	1.684
No vol/VRP	0.772	1.382	0.587	1.600
No skew/tail	0.772	1.372	0.586	1.629
No relative value	0.822	1.355	0.504	1.570

Table 4: Structural-mean channel ablation. Each row re-solves and re-scores the locked E1 book with one expected-return channel set to zero, holding the covariance, caps, and cost stack fixed; cells are full-cost net Sharpe. The Full E1 row reproduces the headline scoreboard by construction and is enforced by the verifier.

Table 4 shows which expected-return channel actually carries each book. The answer is not uniform across universes. For the two VIX-enabled books the carry channel is load-bearing: removing it collapses `orig+VIX` from 1.383 to 0.463 and `larger+VIX` from 1.628 to 0.483, while removing the volatility, skew/tail, or relative-value channels moves those books by at most a few hundredths. The optimizer's VIX edge in this sample is therefore priced premium capture under unchanged state variables, disciplined by the stress and vega budgets, rather than a directional volatility forecast. For the no-VIX eight-name book the ranking reverses: the delta channel is the engine (0.778 falls to 0.370 without it), and carry is nearly irrelevant. At broad no-VIX breadth the relative-value term contributes meaningfully (0.587 versus 0.504 without it).

Two readings of this table are excluded. First, the arms in which a removed channel slightly raises the net Sharpe (dropping delta lifts `orig+VIX` to 1.479 and `larger+VIX` to 1.684) are not promoted as improved candidates: E1 was locked before the validation runs, these are post-hoc diagnostics of the locked object, and differences of this size sit well inside the sampling bands reported in Section 5. Second, the ablation does not claim channel-level attribution of realized P&L; it measures how the selected book changes when the optimizer stops seeing one premium source.

The liquidity cap matters for a different reason. A post-hoc capacity penalty lets the optimizer select positions first and then punishes them after the fact. A hard pre-trade cap changes the feasible set before selection. The net-cap formulation also fixed a real solver pathology: when caps were applied to split long/short legs, the solver could burn offsetting exposure as pseudo-cash and consume good contracts' capacity. Binding $|w_i|$ after netting removed that artifact and left non-degenerate cases unchanged.

The VIX result is economically plausible. Equity options provide single-name convexity, skew, and carry exposure. VIX options add a traded volatility instrument (Whaley, 1993) that can respond directly to market-wide volatility jumps. The empirical result says that this extra state channel matters in the tested sample. It does not say that VIX options are free hedges: they have their own spreads, settlement requirements, and wing-liquidity caveats, and the paper keeps those rows marked as proxied where matched historical CBBO is absent.

The cap result is equally important. The old optimizer could generate attractive gross allocations that were later destroyed by capacity costs. The new formulation moves the constraint into the decision. That changes the selected book, not just the accounting afterward. At $1 million NAV and $X = 0.05$, the optimizer chooses a smaller, cleaner set of trades that the cap vector can support. At larger NAV, the cap vector itself becomes too small to support full deployment, so the correct conclusion is not to increase leverage but to stop promoting the book.

Config	Candidates	Active	Top 5 share	Deployed gross	At cap share	Cap budget
orig	49	47	0.421	0.497	0.064	0.739
orig+VIX	54	54	0.324	0.723	0.093	1.093
larger	262	243	0.233	0.594	0.078	1.160
larger+VIX	267	253	0.296	0.836	0.083	1.515

Table 5: Locked E1 book shape at $1 million NAV and $X = 0.05$. Candidates is the number of eligible contracts, Active the number with nonzero weight, Top 5 share the premium share of the five largest positions, At cap share the fraction of active positions at their liquidity cap, and Cap budget the sum of per-contract caps in NAV units. Deployed gross is cross-checked against the locked robustness ledger by the verifier.

Table 5 quantifies how binding this discipline is. In every universe the optimizer touches nearly every eligible contract—all 54 in `orig+VIX`, and 47 of 49, 243 of 262, and 253 of 267 in the others, the remainder rounding to zero whole contracts. Because whole-contract rounding steps each cap-binding leg just inside its cap, realized utilization clusters near but rarely exactly at the cap: median leg utilization is 0.86–0.94 across the four books, while only 6–9 percent of legs land within one percent of the cap after rounding. Breadth buys diversification of the cap budget rather than concentrated conviction: moving from 49 to 267 candidates lowers the top-five premium share from 0.421 to 0.296 while raising the deployable cap budget from 0.739 to 1.515 NAV units. The book's shape is set jointly by the cap vector and the mean/covariance inputs, not by the covariance alone. The `orig` row also makes the capacity diagnosis concrete: its cap budget of 0.739 NAV is below one, which is exactly why that row is labeled capacity-infeasible despite its positive Sharpe.

4.3. Interpretation by scenario

The `orig` no-VIX row is useful mainly as a control. It uses exact historical CBBO for the eight equity names and achieves a positive point estimate, but it is capacity-infeasible under the hard cap budget. The result therefore validates pieces of the framework without becoming a deployable headline book.

The `orig+VIX` row is a stronger result. It uses exact equity-option CBBO for the eight names and an inferred proxy for the VIX spread rows. It beats stock Markowitz and capped naive options, and its MC refit p05 remains above one Sharpe point. The caveat is execution data for VIX rows, not the basic allocation mechanism.

The `larger` no-VIX row is mixed. The regularization and cap fixes make the result positive, but the optimizer does not beat capped naive options by a meaningful amount and it remains below the stock Markowitz Sharpe. This row is evidence that breadth alone is not enough.

The `larger+VIX` row is the strongest result. It has the best E1 Sharpe, best Sortino, strongest rolling Sharpe, and strongest refit-stability statistics. It also carries the largest spread-imputation dependence, because roughly four-fifths of broad cost rows use the inferred CBBO proxy rather than exact historical CBBO; its performance should be read under the calibrated spread assumptions of Section 3.

4.4. Economic interpretation

The four scenarios separate three mechanisms. The first is option structure: E1 has to beat capped naive options, otherwise the theory adds little. The second is volatility instrument access: adding VIX materially improves both the original and broad universes. The third is breadth: adding names helps only when the estimator is regularized and VIX is present. Without VIX, the broad universe becomes positive but does not clear the baseline hurdle.

The result also shows why option-only strategies should be evaluated by deployable notional rather than by nominal contract count. The broad universe contains many more contracts, but the added liquidity budget is much smaller than the added choice set. That asymmetry creates the central estimation-capacity tradeoff. More names reduce idiosyncratic dependence and create more spread opportunities, but they also increase mean-estimation noise and may not increase executable premium capacity enough.

The strongest practical lesson is that breadth and liquidity caps are complements. A breadth expansion without cap discipline lets the optimizer choose thin options and later pay capacity penalties. A cap without mean regularization can keep trades small but still select noisy expected returns. The positive VIX-enabled rows appear only when both pieces are present.

4.5. What the point estimates do and do not say

The point estimates are economically meaningful because they are full-cost net numbers, but they remain sample estimates. A net Sharpe of 1.628 for `larger+VIX` does not mean the population Sharpe is known to three decimals. It means that, in the realized test path and under the corrected cost stack, the locked E1 book earned a substantially better risk-adjusted return than both baselines. The validation section then asks whether that ordering survives alternate histories and refits, and Table 8 attaches confidence intervals, deflated Sharpe ratios, and formal Sharpe-difference tests to these point estimates.

The large Sortino ratios should be read similarly. Option books can have asymmetric return distributions, so downside semideviation is a useful complement to standard deviation. But Sortino ratios can also look high when the sample has few downside months. That is why the paper does not rely on Sortino alone. It reports rolling OOS, MC lower tails, drawdown breach rates, and CPCV stress tests in the same result package.

The stock Markowitz baseline is not a perfect substitute for the option book because it lacks convexity and volatility exposure. It is nevertheless a hard economic comparison: if an option strategy cannot beat a simple stock portfolio after paying option trading costs, the option-

Config	Full net p05	Full net p50	Default share	Claim net p05	Claim net p50	Rel full p05	Rel claim p05
orig	-0.167	-0.124	1.000	0.437	0.490	0.018	-0.125
orig+VIX	-0.134	-0.115	1.000	1.003	1.062	0.238	2.257
larger	-0.376	-0.278	0.909	0.660	0.706	0.443	-0.152
larger+VIX	-0.454	-0.320	1.000	1.445	1.474	0.426	1.501

Table 6: CPCV path distributions under the two designs. Full columns are recombined-path net Sharpe quantiles when folds span the entire 2015–2026 panel; Default share is the fraction of full-history net paths whose compounded wealth is absorbed at zero, reported separately from the Sharpe statistics. Claim columns restrict the held-out groups to the post-2020 evaluation window. Rel columns are fifth percentiles of per-path Sharpe differences between E1 and the matched capped-naive book on the same folds and costs, under the full-history and claim-window designs respectively.

specific complexity is not justified. The capped naive option baseline is the complementary comparison: it keeps the option asset class but removes the theory. Passing both baselines is why the VIX-enabled rows deserve attention.

The broad no-VIX row is the clearest example of why this distinction matters. It has a positive Sharpe, so it is not a failed strategy in the narrow sense. But it is almost identical to capped naive options and below the stock baseline. A paper that promoted that row as evidence for Greek Markowitz would overstate the finding. Here it is marked mixed and used to show that breadth without a traded volatility instrument is not enough in the current sample.

5. Robustness and distributional checks

5.1. Cross-validation and overfitting controls

The validation layer first asks whether the result depends too much on one chronological split. Combinatorially purged cross-validation is run in two designs that answer two different questions. The *full-history* design divides the entire monthly panel (2015–2026) into 12 contiguous groups; each fold refits the locked E1 procedure on the remaining groups after a one-month purge and embargo, scores the held-out pair, and the $\binom{12}{2} = 66$ splits recombine into 11 complete backtest paths. The *claim-window* design applies the identical machinery, but the 12 groups partition only the post-2020 evaluation months on which every claim in this paper is made; each fold trains on all earlier history plus the non-held-out claim months. The full-history design stresses the procedure far outside its claimed domain; the claim-window design is the correctly adapted test of the claim itself. The PBO summary combines the 12 blocked splits and 66 CPCV splits, for 78 total CV splits.

CPCV of either design is a robustness diagnostic, not a tradable out-of-sample history. Some CPCV training sets include months after the test months, by construction. Its role is to stress the selection and path distribution, estimate probability of backtest overfitting, and reveal whether a result comes from one lucky chronological window. The paper therefore reports CPCV and PBO as validation evidence, while the trading claim remains tied to the locked E1 model and the static/rolling out-of-sample protocols.

The two designs disagree, and the disagreement is informative rather than embarrassing. Full-history CPCV net paths are negative for every variant even though the same paths are solidly positive gross (median gross path Sharpes range from 0.81 to 1.14). The mechanism is concrete: recombined paths force fold-refit books through 2015–2017 and 2020 months in which the corrected $1 million cost stack is at its most punitive relative to the thin early

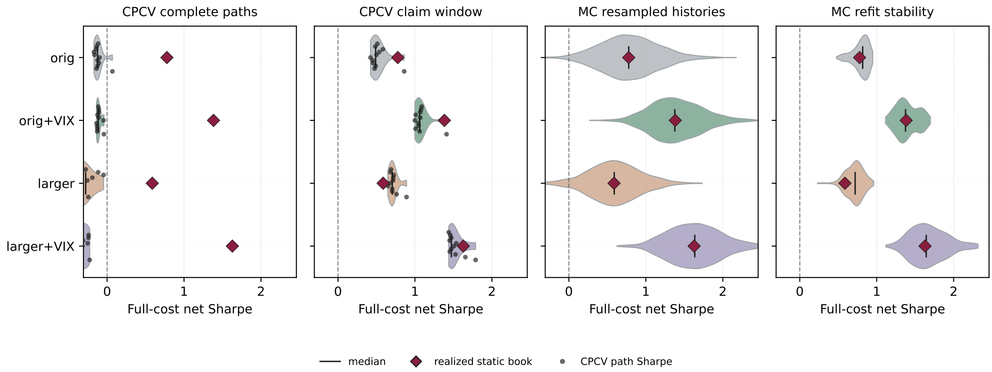

Figure 3: Distributional validation for all four variants. Each shape is the density of full-cost net Sharpe ratios across validation paths (CPCV panels overlay the 11 individual path values as points); black bars mark medians and diamonds mark the realized static-book Sharpe. MC refit and resampled evidence supports the VIX-enabled candidates; full-history CPCV net paths remain a visible stress, while claim-window CPCV paths are positive for all four variants.

panel, producing isolated months with net returns below -100% (24 of 1,419 path-months for `orig+VIX`) that no 129-month compounding survives—nearly every full-history net path is absorbed at zero, versus half of the gross paths. The allocation mechanism survives recombination; what fails is the interaction of a modern cost model with a pre-2021 market the paper never claims to trade. Restricted to the claimed window, the same machinery reverses this verdict: recombined net path distributions are positive for every variant, with fifth-percentile and median path Sharpes of $0.44/0.49$ for `orig`, $1.00/1.06$ for `orig+VIX`, $0.66/0.71$ for `larger`, and $1.45/1.47$ for `larger+VIX` (Table 6).

The relative statistic sharpens this reading. Because the naive baselines are scored on the same folds under the same costs, per-path Sharpe differences between E1 and the matched capped-naive book difference out the common cost drag. Over the full 2015–2026 stress, E1 beats its matched naive book on every recombined path, with fifth-percentile Sharpe advantages between $+0.02$ and $+0.44$: even in the years where both books lose after costs, the optimizer loses less. Inside the claim window the relative paths separate exactly along the paper's verdicts. The VIX-enabled books clear their naive baselines by $+2.26$ and $+1.50$ Sharpe points at the fifth percentile, while the no-VIX books' fold-refit advantages do not clear zero at the fifth percentile (-0.13 and -0.15). The latter is worth stating carefully: the locked static `orig` book does beat its naive baseline (Table 8), but that advantage is not stable under fold-level refitting, which is the CPCV analogue of the mixed and capacity-diagnostic verdicts those rows already carry.

Two features of this analysis bear emphasis. First, the pass/mixed verdicts reported in Section 4 continue to use the harsher full-history CPCV tail as one of their five criteria; the verdict formula was fixed before the claim-window analysis was added and is not re-scored on the friendlier statistic. Second, the full-history stress is retained in every figure and table rather than replaced, because it documents a real property of the strategy class: at this account size, option books with monthly full-cost rebalancing do not survive being transplanted into the thin pre-2021 panel.

The PBO interpretation is similarly bounded. The PBO rank screen asks whether the row

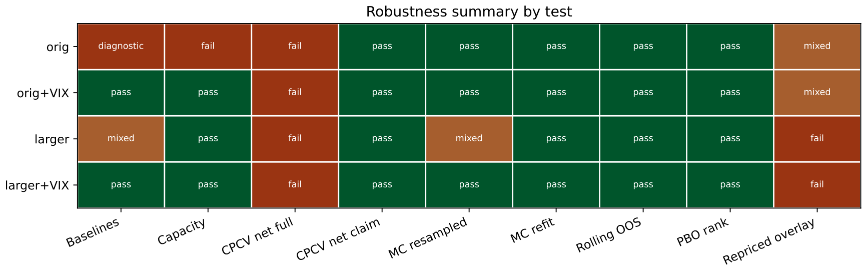

Figure 4: Robustness summary by test. Repriced synthetic paths use a historical-cost overlay rather than synthetic NBBO; the full-history CPCV test is the binding stress, while its claim-window version is positive for all four variants.

chosen in sample tends to rank poorly out of sample across the cross-validation splits. Passing this screen reduces concern that the observed row is merely a backtest winner among many alternatives. It does not eliminate development risk, because the research program itself learned from earlier failed configurations. The paper handles that by locking E1 before the final robustness run and by reporting the no-VIX and proxy-spread caveats plainly.

5.2. *Monte Carlo, repricing, and path risk*

The Monte Carlo layer has three roles. Resampled historical universes preserve the joint cross-sectional strategy-return dependence while changing month order through circular-block sampling. Refit stability resamples training histories, refits the locked procedure, and measures how sensitive the final book is to training-sample variation. Repriced synthetic universes rebuild option paths from simulated state variables and model prices, then apply a documented historical-cost overlay.

The repriced layer is a stress world by construction. It is not synthetic NBBO. Black–Scholes and Black–76 repricing can test exposure to spot, volatility, VIX, and state-path assumptions, but it does not reproduce historical order books, queue priority, displayed size, or market-maker widening. Repriced synthetic paths therefore answer a model-risk question, not an execution-quality question. Tail-path simulation diagnostics are path-risk diagnostics only.

The failure or mixed status of repriced overlay rows should not be over-read. A repriced Black–Scholes/Black–76 world does not contain the same historical option risk premia as the observed market. If the strategy earns compensation from selling or buying specific real-world insurance premia, a synthetic repricing model can remove part of the edge by construction. The useful information is sensitivity: which variants remain stable when the option surface is rebuilt from state paths, and which variants depend heavily on the observed realized surface.

Drawdown breach simulations and GARCH/EGARCH-style volatility diagnostics then ask a different question: how likely is the strategy to violate a risk tolerance under stressed path dynamics? EGARCH is used when the sample is long enough; otherwise the implementation falls back to the simpler volatility-path diagnostic. The paper does not promote those simulations to alternate realized histories. They are scenario tools for risk review.

The robustness matrix supports a narrow conclusion. The VIX-enabled rows pass baseline, capacity, claim-window CPCV, MC resampled, MC refit, rolling OOS, and PBO-rank screens, while failing or mixing on the most conservative repriced-overlay and full-history

Config	Baselines	Capacity	CPCV net full	CPCV net claim	MC resampled	MC refit	Rolling OOS	PBO rank	Repriced overlay
orig	diagnostic	fail	fail	pass	pass	pass	pass	pass	mixed
orig+VIX	pass	pass	fail	pass	pass	pass	pass	pass	mixed
larger	mixed	pass	fail	pass	mixed	pass	pass	pass	fail
larger+VIX	pass	pass	fail	pass	pass	pass	pass	pass	fail

Table 7: Pass/mixed/fail summary by robustness test. "PBO rank" is based on the within-configuration full-cost-net PBO row.

CPCV net tests. The broad no-VIX row is mixed because its realized point estimate is positive but its optimizer edge is negligible and its resampled lower tail crosses zero. The original no-VIX row remains a diagnostic because capacity, not forecast quality, blocks a deployable verdict.

The distributional checks therefore support a ranking rather than a binary sales claim. The first tier is `larger+VIX`, which has the strongest point estimate and the best refit/rolling support. The second tier is `orig+VIX`, which is cleaner on equity spread sourcing but still uses proxied VIX spreads. The third tier is `larger`, which is positive but not meaningfully better than capped naive or stock Markowitz. The fourth tier is `orig`, which is useful for exact-CBBO diagnostics but not deployable under the cap budget.

5.3. Multiple testing and rolling OOS

The multiple-testing layer prevents the paper from treating the best observed row as a free discovery. The reality-check and deflated-Sharpe style diagnostics (White, 2000; Hansen, 2005; Bailey and López de Prado, 2014) account for strategy search and finite-sample selection; HAC-based family inference (Newey and West, 1987) stays in the artifact ledgers. The locked E1 robustness run is designed to reduce this caveat: the final validation does not retune the specification inside test folds. Nevertheless, the model was developed through earlier diagnostics, so the paper reports the remaining selection risk rather than claiming a pristine randomized experiment.

Config	Net Sharpe	CI lo	CI hi	PSR	DSR	dSR stock	p stock	dSR naive	p naive
orig	0.778	0.163	1.433	0.973	0.945	-0.131	0.776	0.413	0.014
orig+VIX	1.383	0.895	1.971	1.000	0.997	0.474	0.383	2.273	0.000
larger	0.587	0.029	1.173	0.925	0.745	-0.322	0.436	-0.009	0.967
larger+VIX	1.628	1.111	2.245	1.000	0.996	0.719	0.179	1.335	0.000

Table 8: Formal inference on the static full-cost net books over the 60 test months. CI lo/hi are 90 percent circular-block bootstrap bands for the net Sharpe. PSR is the probabilistic Sharpe ratio against a zero benchmark and DSR is the deflated Sharpe ratio charged for the 22 optimizer candidates in the locked estimation grid. dSR columns are annualized Sharpe differences of E1 minus the stock Markowitz baseline and minus the matched capped-naive baseline, with two-sided Jobson–Korkie–Memmel p-values.

Table 8 states what the sample can and cannot support. The two VIX-enabled books are individually solid: their 90 percent Sharpe bands exclude zero (0.895–1.971 for `orig+VIX`, 1.111–2.245 for `larger+VIX`), their probabilistic Sharpe ratios both round to 1.000, and their deflated Sharpe ratios remain 0.997 and 0.996 after charging for the 22 candidate specifications examined in the estimation phase (Bailey and López de Prado, 2014). The no-VIX rows are weaker on the same evidence: the `larger` band touches zero at its lower edge and its DSR falls to 0.745.

The Sharpe-difference tests (Jobson and Korkie, 1981; Memmel, 2003), computed on paired monthly series and confirmed by a paired circular-block bootstrap in the spirit of Ledoit and Wolf (2008), draw the sharpest distinction in the paper. Against the capped-naive baselines, the optimizer edge of the VIX-enabled books is large and statistically significant: $+2.273$ Sharpe points for `orig+VIX` and $+1.335$ for `larger+VIX`, both with $p < 0.001$, and $+0.413$ with $p = 0.014$ even for the capacity-infeasible `orig` diagnostic. The `larger` no-VIX edge over naive is essentially zero and slightly negative (-0.009, $p = 0.967$), which is the formal version of the "mixed" verdict. Against stock Markowitz, however, no variant's Sharpe advantage is statistically significant on 60 months: the differences range from -0.322 to $+0.719$ with p-values between 0.179 and 0.776.

This asymmetry belongs in the statement of results, not in a footnote. The finding that survives formal testing is that Greek-induced allocation with VIX access beats naive option diversification after full costs. The finding that the VIX-enabled books beat the stock baseline is a point-estimate ordering that the current sample length cannot separate statistically; a 0.72 annualized Sharpe gap simply requires more than 60 monthly observations to resolve at conventional levels. The paper therefore leans on the baseline comparisons, the rolling OOS refits, and the MC refit distributions jointly rather than on any single test.

Rolling 36-month OOS refits are the most trading-like validation in the current package. They refit the model through time and score subsequent months under the same cost and liquidity rules. The rolling Sharpe values are 1.273 for `larger+VIX`, 1.033 for `orig+VIX`, 0.585 for `orig`, and 0.480 for `larger`. Those numbers reinforce the ordering in the static scoreboard: VIX-enabled E1 performs; no-VIX breadth is not enough.

Figure 5 is the most useful visual bridge from the scoreboard to the validation logic. It shows that the VIX-enabled E1 rows do not win only through one terminal statistic; their cumulative paths recover from the early sample drawdown and then separate after 2024. It also shows why the capped-naive VIX baselines are not acceptable substitutes: the matched naive VIX books decay sharply even though some naive no-VIX paths recover later. The regular stock Markowitz path is a strong baseline, but it does not match the rolling OOS compounding of the two VIX-enabled E1 variants.

5.4. Interpreting the robustness tests

The tests are not averaged into a single meta-score because they answer different questions. Capacity asks whether the target book can be deployed at the specified NAV. Baseline dominance asks whether the option theory beats simpler alternatives. Full-history CPCV asks whether path ordering and fold composition create fragility far outside the claimed period; claim-window CPCV asks the same question inside it. MC resampling asks whether the realized sequence is unusually favorable. MC refit asks whether the procedure is stable to training histories. Repriced overlay asks whether the result depends on the realized option surface and cost path. Rolling OOS asks whether the model can be refit through time without relying on one static split.

The pass/mixed/fail labels should therefore be read qualitatively. A full-history CPCV net failure does not automatically reject the static candidate, because that design is non-tradable, spans a decade the claim does not cover, and is harsh after full costs. A capacity failure does reject deployable status, because a book that cannot be formed at the target NAV is not a trading candidate. A repriced overlay failure is a model-risk warning, not proof that historical edge is illusory. The paper's final ranking follows the tests that are closest to the intended claim: deployability, full-cost baseline dominance, claim-window CPCV, MC refit stability, rolling OOS, and spread-source transparency.

This framework is also why the paper does not chase a perfect all-green heatmap. A stress battery worth reporting should contain visible failures. If every robustness test were passed easily, the likely explanation would be that the tests are too weak. The goal is to identify which limitations remain after the main methodological errors have been fixed.

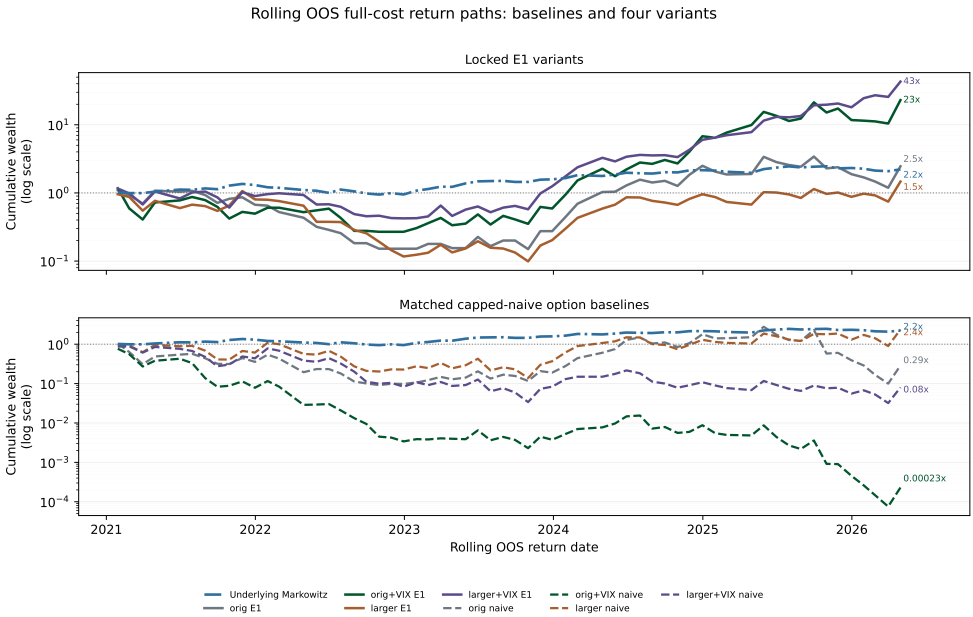

Figure 5: Rolling OOS full-cost cumulative wealth paths. The top panel compares the four locked E1 variants with underlying stock Markowitz. The bottom panel compares the matched capped-naive option baselines with the same stock baseline. The VIX-enabled E1 variants are the only option books that both compound strongly and beat the stock baseline in this walk-forward ledger.

6. Capacity, execution, and limitations

6.1. Capacity and scale

The main practical brake is market depth. At \$1 million NAV, the two VIX-enabled books pass the small-account research test. Above roughly \$2 million, the representative month-end option contracts no longer support full deployment under the tested participation caps. At \$5 million and above, the sum of train-window liquidity caps falls below one NAV for every tested configuration and participation level. Positive Sharpe rows that deploy only a tiny fraction of NAV are economically vacuous and are not promoted.

Breadth increases the cap budget, but not in proportion to the number of underlyings. Moving from eight equity names to 56 names increases the number of candidate contracts by far more than it increases executable month-end premium capacity. Many added-name options

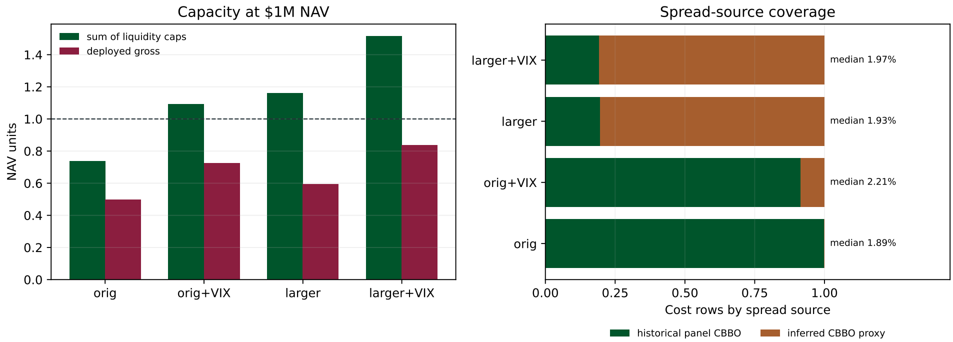

Figure 6: Capacity and spread-source coverage. The left panel shows that deployability is a binding part of the result. The right panel separates exact historical panel CBBO from inferred CBBO proxy rows.

are thin. That is why the paper's positive claim is a small-account claim. The option market, not the conic optimizer, becomes the wall.

This finding also determines how the capacity result should be read for implementation. Sizing should not begin by asking how much capital can be allocated to the best Sharpe row; it should ask what NAV can be deployed while respecting per-contract displayed size, per-underlying concentration, and margin limits. The research answer points to a small-account allocation, not a scalable standalone fund. Additional capital is better spent expanding the execution universe with verified quotes and displayed size than multiplying the same representative contracts.

6.2. Execution caveat

The spread proxy is the largest remaining empirical caveat. It is a major improvement over the old default spread assumptions and stale quote snapshots, but it still cannot replace matched market-hours historical NBBO/CBBO for every broad and VIX row. The correct use of the broad results is therefore calibrated execution sensitivity: if broad large-cap and VIX spreads behave like the liquid CBBO surface used for imputation, locked E1 survives with VIX at \$1 million NAV. The result should not be read as a proof that every historical target contract could have been filled at the imputed quote.

The same distinction applies to margin and assignment. The simulation charges margin drag and assignment penalties, but a simulated penalty is not a broker statement, so execution quality at realized fills is not claimed; establishing it would require market-hours quotes with displayed size, realized fill records, broker margin, and reconciliation against actual statements, all outside the scope of this paper.

Ex-dividend and settlement handling is especially important for options. American single-name options can be assigned early, especially around ex-dividend dates and deep in-the-money short calls. VIX options have their own settlement mechanics and require exact VRO/SOQ handling for expiry P&L. These are not cosmetic accounting details. They can change realized cashflows and margin usage. The simulation therefore prices these events with explicit penalties and settlement labels, and the verification harness requires exact settlement data before VIX rows enter the results.

7. Conclusion

This paper develops a portfolio theory for books composed entirely of listed options and subjects it to an unusually strict empirical protocol. The theoretical contribution treats option positions as premium-funded, expiring cashflows rather than transformed asset returns: premium weights replace the unit budget, covariance is induced by a Greek/state map plus regularized residual risk, expected returns are built from structural risk-premium channels rather than historical contract means, and the allocation is a conic program whose feasible set carries the market frictions—gross and short premium, Greek and stress budgets, margin, and pre-trade net liquidity caps—inside the optimization rather than as after-the-fact accounting. The supporting results formalize why this discipline is necessary: the unconstrained option frontier is a ray with no natural scale, estimation error in contract-level means grows like $\sqrt{\log n/T}$ and reverses the value of historical means at breadth, and representation-dependent constraints can silently destroy capacity.

The empirical findings are stated with their statistical limits. Net of a corrected, contract-level transaction-cost stack, the two VIX-enabled books earn full-cost net Sharpe ratios of 1.628 and 1.383 (gross 2.010 and 1.675) and beat capped naive option diversification by 1.3 to 2.3 Sharpe points with $p < 0.001$; the same books exceed the stock Markowitz baseline in point estimate, but sixty monthly observations cannot resolve that gap at conventional significance levels, and the paper says so rather than claiming otherwise. Channel ablations locate the priced source in option carry disciplined by stress and vega budgets, not in directional volatility forecasts. Cross-validation over the deployment window, Monte Carlo resampling, refit stability, and rolling out-of-sample refits support the ordering, while the full-history cross-validation stress and the repriced no-premium worlds mark its limits. Capacity is a first-class result rather than a footnote: displayed option depth supports the strategy at roughly one million dollars of NAV and refuses it at five, so the finding is a statement about small-account option allocation, not about a scalable fund.

Three limitations define the research frontier. Exact market-hours quote history for the broad and VIX contracts would replace the calibrated spread proxy with execution evidence. Longer samples would sharpen the baseline comparisons that sixty months leave open. And richer volatility instruments—VIX futures spreads, variance swaps, or options on other volatility indices—would test whether the carry-channel result generalizes beyond the single traded volatility surface used here. Within its stated scope, the paper's answer to its opening question is affirmative: option structure, priced and constrained correctly, earns its complexity.

A. Technical appendix

The main text keeps the proof sketches close to the economic argument. This appendix records the two mathematical details from the longer draft that are still useful for a research-grade version of the paper: why the option-only optimizer has a well-defined scale only after operational budgets are imposed, and how the maximum-Sharpe solver is connected to a convex conic program.

A.1. Scale, costs, and no-free-exposure logic

Proposition A.1 (No free option exposure)**.** *Let $\mathcal{C}$ include binding gross-premium, short-premium, stress-loss, margin/collateral, or contract-level liquidity controls. Then no economically nonzero sequence of option positions can take unbounded premium, payoff, or Greek exposure while remaining feasible and using zero capital.*

Proof. The listed controls are positively homogeneous in the premium-weight vector. If $\|w_k\| \to \infty$ along an economically nonzero direction, at least one controlled premium, margin, stress, Greek, or liquidity functional diverges. A direction in the null space of all such functionals has no priced premium exposure and no admissible tradable contract size in this model. Therefore every economically meaningful option direction must consume a real budget before scale can diverge. □

This proposition is the formal reason the paper does not treat gross option Sharpe as a trading claim. A cheap option can carry large return exposure per dollar of premium, and a short option can receive premium while creating a future liability. The optimizer is only an allocation rule after premium, short-premium, stress, margin, concentration, and liquidity limits turn the scale-free return object into a feasible book.

Proposition A.2 (Operational monotonicity)**.** *For*

$$V(\mathcal{C}, c) = \max_{w \in \mathcal{C}} \widehat{\mu}^\top w - \frac{\gamma}{2} w^\top \Sigma w - c^\top |w|,$$

with $\Sigma \succeq 0$, if $\mathcal{C}_1 \subseteq \mathcal{C}_0$ and $c_1 \geq c_0$ componentwise, then $V(\mathcal{C}_1, c_1) \leq V(\mathcal{C}_0, c_0)$.

Proof. For every $w \in \mathcal{C}_1$, the same w is feasible for $\mathcal{C}_0$, and $-c_1^\top |w| \leq -c_0^\top |w|$. Maximizing a weakly lower objective over a weakly smaller feasible set cannot increase the value. □

The monotonicity result justifies the conservative reporting convention. Higher spread, fee, slippage, margin, assignment, or impact assumptions can only make the utility value worse. Tighter displayed-size, stress, or margin budgets can only make the feasible set smaller. Positive results under full costs and hard caps are therefore more meaningful than positive gross results, while failures under exact execution data would supersede the current proxy-spread sensitivity.

A.2. Mean estimation error, shrinkage, and purge details

Proof of Proposition 2.4. Each error e_i is sub-Gaussian with variance proxy $\sigma_i^2/T \leq \bar{\sigma}^2/T$, so for every $\lambda > 0$, $\mathbb{E} \exp(\lambda e_i) \leq \exp(\lambda^2 \bar{\sigma}^2 / 2T)$ and the same holds for $-e_i$. Let $M =$

$\max_i |e_i|$, the maximum of the $m = 2n$ variables $\{\pm e_i\}$. By Jensen's inequality and a union bound on the moment generating function,

$$\exp\big(\lambda\,\mathbb{E}[M]\big) \le \mathbb{E}\exp(\lambda M) \le \sum_{j=1}^{m} \mathbb{E}\exp(\lambda X_j) \le m\exp\Big(\frac{\lambda^2\bar{\sigma}^2}{2T}\Big),$$

so $\mathbb{E}[M] \le \log(m)/\lambda + \lambda\bar{\sigma}^2/2T$. Optimizing at $\lambda = \sqrt{2T\log m}/\bar{\sigma}$ gives $\mathbb{E}[M] \le \bar{\sigma}\sqrt{2\log(2n)/T}$, the first display. For the second, Hölder's inequality gives $\sup_{w\in\mathcal{C}} e^\top w \le \sup_{\|w\|_1\le g} e^\top w = g\|e\|_\infty = gM$; take expectations. For the objective-loss statement, write the optimizer's concave criterion as $F_\mu(w) = \mu^\top w - R(w)$, where R collects the risk and cost penalties, and let $\widetilde{w}$ maximize $F_{\widehat{\mu}}$ and w^* maximize F_μ over $\mathcal{C}$. Then

$$F_\mu(\widetilde{w}) \;\ge\; F_{\widehat{\mu}}(\widetilde{w}) - g\|e\|_\infty \;\ge\; F_{\widehat{\mu}}(w^*) - g\|e\|_\infty \;\ge\; F_\mu(w^*) - 2g\|e\|_\infty,$$

so the expected sacrifice of true objective value is at most $2g\,\mathbb{E}\|e\|_\infty \le 2g\bar{\sigma}\sqrt{2\log(2n)/T}$. ☐

Proof of Corollary 2.5. The sandwich argument above holds verbatim when the mean input is the structural forecast: with deterministic bias vector $b = \mu^{\text{struct}} - \mu$ and $\|b\|_\infty \le \beta$, the loss bound is $2g\beta$. Comparing the two worst-case bounds, the historical mean has the smaller guarantee exactly when $\bar{\sigma}\sqrt{2\log(2n)/T} < \beta$, which rearranges to $2\log(2n) < T\beta^2/\bar{\sigma}^2$, i.e. $n < \frac{1}{2}\exp(T\beta^2/2\bar{\sigma}^2)$. At fixed T the left side grows in n while the right side is constant, which yields the stated crossover. ☐

The bounds compare worst-case guarantees, not realized losses, and the sub-Gaussian model ignores the correlation and missingness of real option panels; both simplifications favor the historical mean, so the crossover argument is conservative in the direction the paper uses it.

Lemma A.3 (One-month purge suffices for monthly option labels)**.** *Index each label by its return month-end. Assume every tested contract's information window spans at most Δ calendar days and ends at its return month-end, where $\Delta \approx 44$ days is the maximum decision-to-expiry span of the monthly-tenor contracts. If the realized calendar gap between the latest training return month-end and the earliest test return month-end is $G > \Delta$ in every fold, then no training label's information window overlaps any test label's window.*

Proof. A training label with return month-end t_r has window contained in $[t_r - \Delta,\ t_r]$. A test label with return month-end t_e has window contained in $[t_e - \Delta,\ t_e]$, which begins at $t_e - \Delta$. Overlap requires $t_r > t_e - \Delta$, i.e. $t_e - t_r < \Delta$. The purge/embargo schedule enforces $t_e - t_r \ge G > \Delta$ for every training/test pair, so the windows are disjoint. ☐

The quantity G is not assumed; it is measured. The verifier recomputes the minimum gap across all folds, schemes, and configurations from the fold schedule and ledger artifacts and asserts $G > 44$ days; in the current artifacts the realized minimum is $G = 59$ days. The one-month purge plus one-month embargo is therefore sufficient with two weeks of slack, and the sufficiency claim is re-audited on every verification run rather than hand-typed.

A.3. Maximum Sharpe and conic reduction

Let $\mu = \widehat{\mu}_t$ and $\Sigma = \widehat{\Sigma}_{O,t}$. When $\Sigma \succ 0$, the unconstrained maximum-Sharpe direction is the usual tangency direction even though the assets are option cashflows:

$$\arg\max_{w \neq 0} \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}} \quad \propto \quad \Sigma^{-1}\mu.$$

The difference from stock Markowitz is not the algebra of the tangency direction; it is the construction of μ, Σ, and the feasible set. Option premium weights have cash and collateral as the numeraire, so the unit-stock-budget equation is replaced by gross premium, short-premium, stress, margin, and liquidity budgets.

Proposition A.4 (Conic maximum-Sharpe form)**.** *Let $K \subseteq \mathbb{R}^n$ be a closed convex cone containing some w with $w^\top \mu > 0$, and let $\Sigma \succ 0$. Then*

$$\sup_{w \in K \setminus \{0\}} \frac{w^\top \mu}{\sqrt{w^\top \Sigma w}} = \max\{\, y^\top \mu : \; y \in K, \; y^\top \Sigma y \leq 1 \,\}.$$

The right side is a convex quadratically constrained program and is a second-order-cone program when a Cholesky or square-root factor of Σ is used.

Proof. For any nonzero $w \in K$, define $y = w/\sqrt{w^\top \Sigma w}$. Since K is a cone, $y \in K$, $y^\top \Sigma y = 1$, and $y^\top \mu$ equals the Sharpe ratio of w. This shows that the right side is at least the left. Conversely, any feasible y with $y^\top \mu > 0$ satisfies

$$\frac{y^\top \mu}{\sqrt{y^\top \Sigma y}} \geq y^\top \mu,$$

so the left side is at least the right. Positive definiteness makes the risk ball bounded, and the objective is linear on a closed convex feasible set. □

The production-candidate problem in the paper has bounded budgets rather than a pure cone, so implementation uses the same geometry with a gross-risk or gross-premium normalization and then imposes the operational caps directly. Absolute costs are handled by positive and negative parts, $w = w^+ - w^-$, but the liquidity cap is imposed on the net economic variable:

$$-\bar{w}_{i,t} \leq w_i^+ - w_i^- \leq \bar{w}_{i,t}.$$

Imposing the cap separately on w_i^+ and w_i^- is not equivalent. It allows offsetting positive legs to satisfy an algebraic representation while consuming displayed liquidity without taking the intended net contract exposure.

A.4. Risk-model estimator repair

The population covariance identity in Proposition 2.3 is positive semidefinite by construction. The finite-sample estimate can still fail PSD in one practical place: residual covariance estimated from pairwise-complete option residuals on an unbalanced contract panel. The implementation therefore applies the following repair before the matrix can enter an optimizer:

$$\widehat{\Sigma}_{\varepsilon,t} = Q\Lambda Q^\top \quad \Longrightarrow \quad \widehat{\Sigma}^{+}_{\varepsilon,t} = Q \max(\Lambda, \lambda_{\text{min}} I) Q^\top,$$

with a small nonnegative eigenvalue floor. The factor term $\widehat{B}_t \widehat{\Omega}_t \widehat{B}_t^\top$ is PSD whenever $\widehat{\Omega}_t \succeq 0$, so the repair is targeted to the residual block rather than used as a blanket way to hide a bad risk model.

E1 then makes one additional research choice: it keeps the residual covariance diagonal. That is not a statement that residual option risks are literally independent. It is a regularization choice motivated by the breadth experiment. In the broad universe, dense residual covariance estimated from short, changing contract panels became a false diversification channel. Diagonal residual risk preserves contract-specific unexplained variance while refusing to invert noisy off-diagonal residual moments.

A.5. Appendix use and audit trail

Mathematical mechanics that strengthen the results but would slow the main reading path are kept in this appendix. Large empirical ledgers remain machine-readable artifacts under `analysis/artifacts/breadth_solutions/robustness/`; the body prints only the scoreboard, distributional summaries, robustness heatmap, capacity/spread panel, and rolling OOS paths. Every number in the body traces to one of those ledgers, and the verification harness recomputes the headline statistics from the artifacts on each build.

Data availability statement

The reproduction package contains code, generated tables, figures, verification outputs, schemas, and artifact hashes. Raw licensed OPRA/Databento data are not redistributed. The baseline eight equity-option spread rows use local historical panel CBBO. Broad-name and VIX spread rows that lack matched historical CBBO use the documented inferred CBBO proxy and are labeled as execution-sensitivity evidence. Exact VRO/SOQ settlement files are versioned by source hash before VIX option rows enter the reported tables.

References

D. H. Bailey and M. López de Prado. The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality. *The Journal of Portfolio Management*, 40 (5):94–107, 2014.

D. H. Bailey, J. M. Borwein, M. López de Prado, and Q. J. Zhu. The probability of backtest overfitting. *Journal of Computational Finance*, 20(4):39–69, 2017.

G. Bakshi and N. Kapadia. Delta-hedged gains and the negative market volatility risk premium. *The Review of Financial Studies*, 16(2):527–566, 2003.

A. Ben-Tal and A. Nemirovski. Robust convex optimization. *Mathematics of Operations Research*, 23(4):769–805, 1998.

F. Black. The pricing of commodity contracts. *Journal of Financial Economics*, 3(1–2): 167–179, 1976.

F. Black and M. Scholes. The pricing of options and corporate liabilities. *Journal of Political Economy*, 81(3):637–654, 1973.

T. Bollerslev, G. Tauchen, and H. Zhou. Expected stock returns and variance risk premia. *The Review of Financial Studies*, 22(11):4463–4492, 2009.

M. Broadie, M. Chernov, and M. Johannes. Understanding index option returns. *The Review of Financial Studies*, 22(11):4493–4529, 2009.

P. Carr and L. Wu. Variance risk premiums. *The Review of Financial Studies*, 22(3): 1311–1341, 2009.

J. D. Coval and T. Shumway. Expected option returns. *The Journal of Finance*, 56(3): 983–1009, 2001.

V. DeMiguel, L. Garlappi, and R. Uppal. Optimal versus naive diversification: How inefficient is the 1/N portfolio strategy? *The Review of Financial Studies*, 22(5):1915–1953, 2009.

J. Driessen and P. Maenhout. An empirical portfolio perspective on option pricing anomalies. *Review of Finance*, 11(4):561–603, 2007.

J. A. Faias and P. Santa-Clara. Optimal option portfolios. *Journal of Financial and Quantitative Analysis*, 52(1):277–303, 2017.

A. Frazzini and L. H. Pedersen. Embedded leverage. *The Review of Asset Pricing Studies*, 12 (1):1–52, 2022.

N. Gârleanu and L. H. Pedersen. Dynamic trading with predictable returns and transaction costs. *The Journal of Finance*, 68(6):2309–2340, 2013.

D. Goldfarb and G. Iyengar. Robust portfolio selection problems. *Mathematics of Operations Research*, 28(1):1–38, 2003.

R. C. Grinold and R. N. Kahn. *Active Portfolio Management*. McGraw-Hill, 2 edition, 2000.

P. R. Hansen. A test for superior predictive ability. *Journal of Business & Economic Statistics*, 23(4):365–380, 2005.

J. D. Jobson and B. M. Korkie. Performance hypothesis testing with the Sharpe and Treynor measures. *The Journal of Finance*, 36(4):889–908, 1981.

C. Keating and W. F. Shadwick. A universal performance measure. *Journal of Performance Measurement*, 6(3):59–84, 2002.

O. Ledoit and M. Wolf. Honey, I shrunk the sample covariance matrix. *The Journal of Portfolio Management*, 30(4):110–119, 2004.

O. Ledoit and M. Wolf. Robust performance hypothesis testing with the Sharpe ratio. *Journal of Empirical Finance*, 15(5):850–859, 2008.

A. W. Lo. The statistics of Sharpe ratios. *Financial Analysts Journal*, 58(4):36–52, 2002.

M. López de Prado. *Advances in Financial Machine Learning*. Wiley, 2018.

M. Magdon-Ismail and A. F. Atiya. Maximum drawdown. *Risk Magazine*, 17(10):99–102, 2004.

H. Markowitz. Portfolio selection. *The Journal of Finance*, 7(1):77–91, 1952.

C. Memmel. Performance hypothesis testing with the Sharpe ratio. *Finance Letters*, 1(1): 21–23, 2003.

R. C. Merton. Theory of rational option pricing. *The Bell Journal of Economics and Management Science*, 4(1):141–183, 1973.

D. B. Nelson. Conditional heteroskedasticity in asset returns: A new approach. *Econometrica*, 59(2):347–370, 1991.

W. K. Newey and K. D. West. A simple, positive semi-definite, heteroskedasticity and autocorrelation consistent covariance matrix. *Econometrica*, 55(3):703–708, 1987.

D. N. Politis and J. P. Romano. The stationary bootstrap. *Journal of the American Statistical Association*, 89(428):1303–1313, 1994.

W. F. Sharpe. Mutual fund performance. *The Journal of Business*, 39(1):119–138, 1966.

F. A. Sortino and L. N. Price. Performance measurement in a downside risk framework. *The Journal of Investing*, 3(3):59–64, 1994.

R. E. Whaley. Derivatives on market volatility: Hedging tools long overdue. *The Journal of Derivatives*, 1(1):71–84, 1993.

H. White. A reality check for data snooping. *Econometrica*, 68(5):1097–1126, 2000.